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Profits, Scale Economies, and the Gains from Trade and Industrial Policy

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1 Big picture

The paper asks: *how do profits / markups and scale economies change (i) the gains from trade, and (ii) the design of trade policy versus industrial policy?* The key idea is that once industries differ in *markups* (restricted entry) or *scale externalities* (free entry), there is an **internal allocation wedge** across industries in addition to the usual **terms-of-trade (ToT) wedge** across countries.

- **First-best (global efficiency):** use *industrial policy* (a Pigouvian output subsidy) to correct the internal wedge; there is no reason to use trade taxes in the cooperative/global optimum.
- **Unilateral incentives:** a government dislikes that the Pigouvian subsidy expands its exports and worsens its ToT. So unilateral governments under-implement (or even reverse) efficient industrial policy when ToT concerns dominate.
- **Deep vs. shallow agreements:** shallow trade agreements (limits on tariffs) mainly prevent ToT-driven trade wars; deep agreements that also coordinate industrial policy unlock additional gains by allowing countries to implement corrective industrial policies without suffering ToT losses.

Quantitatively, using a multi-country multi-industry input–output model calibrated to WIOD, the paper finds (i) substantial losses from trade-war retaliation, and (ii) large additional gains from *coordinating* industrial policies—especially because estimated industry-level trade elasticities and scale elasticities are **negatively correlated**, which makes unilateral industrial policy particularly tempting to *avoid* (immiserizing growth logic).

2 Notation table from the paper (Table 1)

Table 1 is essentially a notation/objects summary (demand, prices, shares, elasticities, and policy instruments). I treat it as the dictionary for the rest of the paper.

3 Model 1: “Generalized Krugman” multi-country, multi-industry equilibrium

3.1 Preferences and within-/between-country CES structure

Countries are indexed by $i, j \in \mathcal{C}$ and industries by $k \in \mathcal{K}$.

Top-level across industries. Country i has an aggregate utility/consumption aggregator (kept fairly general in the theory; in the quantitative work it is effectively Cobb–Douglas across k).

Within an industry: national varieties. For each industry k , country i consumes a CES aggregate across *national* varieties:

$$Q_{i,k} = \left(\sum_{j \in \mathcal{C}} Q_{ji,k}^{\frac{\sigma_k - 1}{\sigma_k}} \right)^{\frac{\sigma_k}{\sigma_k - 1}}, \quad \sigma_k > 1. \quad (1)$$

Here σ_k governs the *trade elasticity / substitution across countries* within industry k .

Within a national variety: firm varieties. The national variety $Q_{ji,k}$ is itself CES over individual varieties ω produced in j :

$$Q_{ji,k} = \left(\int_{\omega \in \Omega_{j,k}} q_{ji,k}(\omega)^{\frac{\gamma_k-1}{\gamma_k}} d\omega \right)^{\frac{\gamma_k}{\gamma_k-1}}, \quad \gamma_k > 1. \quad (2)$$

γ_k governs *within-country substitution across firm varieties*.

3.2 Price indices and demand shares

Let $\tilde{p}_{ji,k}(\omega)$ denote the *consumer* price (inclusive of policy wedges) in i of variety ω from j in industry k . Define consumer price indices:

$$\tilde{P}_{ji,k} = \left(\int_{\omega \in \Omega_{j,k}} \tilde{p}_{ji,k}(\omega)^{1-\gamma_k} d\omega \right)^{\frac{1}{1-\gamma_k}}, \quad (3)$$

$$\tilde{P}_{i,k} = \left(\sum_{j \in \mathcal{C}} \tilde{P}_{ji,k}^{1-\sigma_k} \right)^{\frac{1}{1-\sigma_k}}. \quad (4)$$

Standard CES demand implies expenditure shares:

$$\lambda_{ji,k} \equiv \frac{\tilde{P}_{ji,k}^{1-\sigma_k}}{\sum_{n \in \mathcal{C}} \tilde{P}_{ni,k}^{1-\sigma_k}} = \left(\frac{\tilde{P}_{ji,k}}{\tilde{P}_{i,k}} \right)^{1-\sigma_k}, \quad (5)$$

$$\lambda_{ji,k}(\omega) \equiv \frac{\tilde{p}_{ji,k}(\omega)^{1-\gamma_k}}{\int_{\omega' \in \Omega_{j,k}} \tilde{p}_{ji,k}(\omega')^{1-\gamma_k} d\omega'} = \left(\frac{\tilde{p}_{ji,k}(\omega)}{\tilde{P}_{ji,k}} \right)^{1-\gamma_k}, \quad (6)$$

(up to any quality shifters; the estimation section makes the quality term explicit).

3.3 Demand elasticities (objects used in the optimal-policy formulas)

The paper works with Marshallian elasticities of (national-variety) demand in market i . A useful special case (Cobb–Douglas across industries) gives for $j \neq i$ in the same industry k :

$$\epsilon_{ji,k}^{(ji,k)} = -1 - (\sigma_k - 1)(1 - \lambda_{ji,k}), \quad (7)$$

$$\epsilon_{ji,k}^{(ji,k)} = (\sigma_k - 1)\lambda_{ji,k}. \quad (8)$$

I read $\epsilon_{ji,k}^{(ji,k)}$ as the (negative) own-price elasticity of national-variety demand in market i ; it combines a unit elasticity from the Cobb–Douglas expenditure effect (the -1 term) plus the CES substitution effect.

For the policy theorems, it is convenient to stack these elasticities into matrices:

$$E_{ij} \equiv \left(\epsilon_{ni,g}^{(ji,k)} \right)_{g,k \in \mathcal{K}},$$

with additional block-notation for submatrices used in Theorems 2–3 (I follow the paper’s notation).

3.4 Production, profits, and scale economies

Let $M_{i,k}$ denote the mass of operating firms (varieties) in country i , industry k . Iceberg trade costs are $\tau_{ij,k}$.

The producer-price index for shipments from i to j in industry k has the “love-of-variety”/scale form (paper’s Eq. (4)):

$$P_{ij,k} = \frac{\gamma_k}{\gamma_k - 1} \tau_{ij,k} \bar{a}_{i,k} w_i M_{i,k}^{-\frac{1}{\gamma_k - 1}}. \quad (9)$$

Define the key reduced-form object

$$\mu_k \equiv \frac{1}{\gamma_k - 1}, \quad \text{so } \frac{\gamma_k}{\gamma_k - 1} = 1 + \mu_k. \quad (10)$$

μ_k is central: it can be interpreted as a *markup/profit wedge* (restricted entry) or as a *scale elasticity* (free entry). The paper explicitly analyzes both cases.

Restricted entry (profits/markups). $M_{i,k}$ is fixed, so prices do not adjust through entry. Firms earn profits. The economy-wide (revenue-weighted) average “profit margin” is (paper’s Eq. (5)):

$$\bar{\mu}_i = \frac{\sum_k \sum_j \frac{\mu_k}{1 + \mu_k} P_{ij,k} Q_{ij,k}}{\sum_k \sum_j \frac{1}{1 + \mu_k} P_{ij,k} Q_{ij,k}}, \quad \text{and (in this case) } \bar{\mu}_i > 0. \quad (11)$$

Define a “profit-adjusted” wage $w'_i \equiv (1 + \bar{\mu}_i) w_i$.

Free entry (scale economies). Entry drives profits to zero, implying $\bar{\mu}_i = 0$ and $w'_i = w_i$, but the mass of varieties $M_{i,k}$ becomes endogenous and generates a scale externality.

Reduced-form price representation (paper’s Eq. (6)). The paper shows that the producer-price indices can be written in convenient reduced form:

$$P_{ij,k} = \begin{cases} \varrho_{ij,k} \left(\sum_{n \in \mathcal{C}} \tau_{in,k} Q_{in,k} \right)^{-\frac{\mu_k}{1 + \mu_k}} w'_i, & \text{free entry (scale effects),} \\ \varrho'_{ij,k} \left(\frac{1 + \mu_k}{1 + \bar{\mu}_i} \right) w'_i, & \text{restricted entry (profits).} \end{cases} \quad (12)$$

3.5 Policy instruments and wedges

Countries have three ad valorem instruments:

- import taxes (tariffs) $t_{ji,k}$ imposed by destination i on imports from j in industry k ,
- export subsidies $x_{ij,k}$ chosen by origin i on exports to j in industry k ,
- industrial (output) subsidies $s_{i,k}$ applied to all output produced in i in industry k .

They enter consumer prices as (paper’s Eq. (7)):

$$\tilde{P}_{ji,k} = \frac{1 + t_{ji,k}}{(1 + x_{ji,k})(1 + s_{j,k})} P_{ji,k}. \quad (13)$$

Lemma 1 in the paper shows a *tax-neutrality* property: some common scalings of (t, x, s) leave allocations unchanged (a normalization issue).

4 Model 2: Efficient allocation and the role of industrial policy

Let $W_i(T)$ denote welfare in i under a feasible policy profile T (paper's Eq. (8)):

$$W_i(T) = V_i\left(Y_i(T), \tilde{P}_i(T)\right).$$

Global first-best (planner). In the cooperative/global optimum, ToT motives vanish. The key result is (paper's Eq. (9)):

$$t_{ji,k}^* = 0, \quad x_{ij,k}^* = 0, \quad 1 + s_{i,k}^* = 1 + \mu_k \quad \forall i, k. \quad (14)$$

Interpretation: the only wedge that needs correction is the within-country scale/markup distortion; the Pigouvian output subsidy corrects it industry-by-industry.

5 Model 3: Unilateral optimal policy (sufficient-statistics characterization)

A central object is the *conditional inverse export supply elasticity* $\omega_{ji,k}$ faced by country i when importing from j in industry k . Theorem 1 shows that optimal unilateral policies take simple sufficient-statistics form.

5.1 Theorem 1 (“first-best” unilateral policy with trade & industrial instruments)

Theorem 1 gives:

$$1 + t_{ji,k}^* = (1 + \bar{t}_i) (1 + \omega_{ji,k}), \quad (15)$$

$$1 + x_{ij,k}^* = \frac{1}{1 + \omega_{ij,k}}, \quad (16)$$

$$1 + s_{i,k}^* = (1 + \mu_k) \frac{1 + \omega_{ii,k}}{1 + \bar{t}_i}. \quad (17)$$

The same theorem provides (paper's Eq. (10)) a simple approximation for $\omega_{ji,k}$:

Restricted entry (profits):

$$\omega_{ji,k} = \left(\frac{1 + \mu_k}{1 + \bar{\mu}_i} \right) \frac{\epsilon_{ji,k}}{\epsilon_{ii,k}}. \quad (18)$$

Free entry (scale economies):

$$\omega_{ji,k} = \frac{\epsilon_{ji,k}}{(1 + \mu_k)(\epsilon_{ii,k} + 1)}. \quad (19)$$

Reading:

- The tariff formula is the standard optimal-tariff structure: import taxes rise with inverse export supply elasticity $\omega_{ji,k}$.
- Export subsidies are the mirror image: subsidize exports more when foreign demand is inelastic (high $\omega_{ij,k}$) to manipulate ToT.
- The industrial subsidy is a Pigouvian term $(1 + \mu_k)$ scaled by general-equilibrium incidence through $(1 + \omega_{ii,k})/(1 + \bar{t}_i)$.

5.2 Theorem 2: second-best trade taxes when industrial policy is unavailable ($s_{i,k} = 0$)

When governments cannot use $s_{i,k}$, trade taxes partially substitute for industrial policy. Theorem 2 characterizes optimal import taxes and export subsidies in vector form. Let $\Omega_{ji} \equiv (\omega_{ji,k})_{k \in \mathcal{K}}$ and let $\mathbf{1}$ be a vector of ones. Using Hadamard operations $\odot$ (elementwise product) and $\oslash$ (elementwise division), the theorem gives:

$$\mathbf{1} + t_{ji}^{**} = (1 + \bar{t}_i) (\mathbf{1} + \Omega_{ji}) \oslash \left[\mathbf{1} + E_{-ii}^{-1} E_{-ii}^{(ii)} (\mathbf{1} - (1 + \mu_k)/(1 + \bar{\mu}_i))_{k \in \mathcal{K}} \right], \quad (20)$$

$$\mathbf{1} + x_{ij}^{**} = -(1 + \bar{t}_i) \left[E_{ij}^{-1} E_{ij}^{(-ij)} (\mathbf{1} + \Omega_{-ii}) \right] \odot ((1 + \mu_k)/(1 + \bar{\mu}_i))_{k \in \mathcal{K}}. \quad (21)$$

Interpretation: without industrial subsidies, optimal trade taxes combine (i) a ToT component $(\mathbf{1} + \Omega_{ji})$ and (ii) an “allocative-efficiency correction” component that depends on cross-industry demand elasticities (the E matrices) and the dispersion of $(1 + \mu_k)/(1 + \bar{\mu}_i)$ across industries.

5.3 Theorem 3: third-best import tariffs when export subsidies are also unavailable ($s_{i,k} = x_{ij,k} = 0$)

With only import taxes available, Theorem 3 yields:

$$\mathbf{1} + t_{ji}^{***} = (1 + \bar{t}_i) \mathbf{1} \oslash \left[\mathbf{1} + E_{-ii}^{-1} E_{-ii}^{(ii)} (\mathbf{1} - (1 + \mu_k)/(1 + \bar{\mu}_i))_{k \in \mathcal{K}} \right]. \quad (22)$$

Interpretation: with tariffs alone, policy tries to reallocate demand across industries toward high- $(1 + \mu_k)$ sectors, but it can do so only indirectly through import wedges (and therefore is a weaker substitute for true industrial policy).

6 Industrial policy implementation and the case for “deep” cooperation

6.1 Figure 1: tension between ToT manipulation and allocative efficiency

The paper emphasizes that trade and industrial policy motives can point in opposite directions depending on how μ_k and σ_k co-move across industries. Figure 1 provides a numerical illustration: the gains from second-best trade policies and the welfare effect of unilateral “markup/scale correction” depend strongly on $\text{cov}(\mu_k, \sigma_k)$.

Takeaway from Fig. 1:

- When $\text{cov}(\mu_k, \sigma_k) < 0$ (high scale/markup sectors are also low trade-elasticity sectors), unilateral correction expands precisely the sectors where ToT losses are largest, so unilateral “efficient” industrial policy can *lower* welfare (immiserizing-growth logic).
- This generates a natural coordination problem: everyone would like industrial distortions corrected globally, but each government has an incentive to hold back unilaterally to free-ride on others’ ToT concessions.

6.2 Table 2: a simple industrial policy coordination game

Table 2 formalizes the implementation problem as a 2×2 game (implement the corrective subsidy $s = \mu$ or withhold). When $\text{cov}(\mu_k, \sigma_k) < 0$, “withhold” can be a dominant strategy even though joint implementation is Pareto-improving.

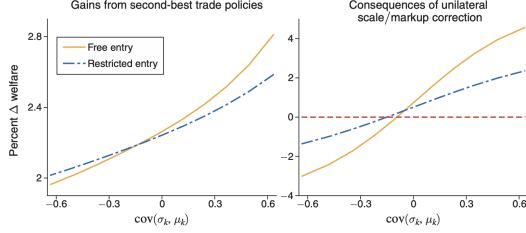


FIGURE 1. TENSION BETWEEN TOT AND ALLOCATIVE EFFICIENCY, WHEN $\text{cov}(\sigma_k, \mu_k) < 0$, CAN YIELD DIRE POLICY OUTCOMES

Notes: This figure corresponds to a two-country and two-industry model with symmetric countries and Cobb-Douglas preferences across industries. The left panel reports the gains from second-best trade taxes, specified by Theorem 2. The trade elasticities in industries 1 and 2 are assigned values $\sigma_1 = 1.5$ and $\sigma_2 = 3$, and scale elasticities are adjusted to vary $\text{cov}(\sigma_k, \mu_k)$. The right panel reports the welfare consequences of unilateral scale or markup correction. Industry-level scale elasticities are assigned values $\mu_1 = 0.5$ and $\mu_2 = 0.2$, and trade elasticities are adjusted to vary $\text{cov}(\sigma_k, \mu_k)$.

TABLE 2—THE INDUSTRIAL POLICY IMPLEMENTATION GAME FACING COOPERATIVE GOVERNMENTS WHEN $\text{cov}(\sigma_k, \mu_k) < 0$

		Country j (%Δ W_j)	
		$s_j = 0$	$s_j = \mu$
Country i (%Δ W_i)	$s_i = 0$	(0%, 0%)	(3.7%, -1.2%)
	$s_i = \mu$	(-1.2%, 3.7%)	(2.7%, 2.7%)

Notes: This table corresponds to a two-country and two-industry model with symmetric countries and CES preferences across industries with substitution elasticity 1.2. Industry-level trade elasticities are $\sigma_1 = 1.5$ and $\sigma_2 = 3$, and scale elasticities are $\mu_1 = 0.2$ and $\mu_2 = 0.5$, implying $\text{cov}(\sigma_k, \mu_k) \approx -0.225$. The unique Nash equilibrium is a race to the bottom, wherein $s_1 = s_2 = 0$.

7 Model 4: Extensions (Section IV)

7.1 Mapping to other canonical trade models

The “generalized Krugman” setup is a reduced-form umbrella:

- **Eaton–Kortum with externalities:** can be represented by appropriate mappings into σ_k and an effective μ_k that captures Marshallian externalities.
- **Armington with externalities:** similarly nests with an externality-driven μ_k .
- **Melitz/Pareto selection:** firm selection implies that an output subsidy changes the mass of operating varieties; the paper shows that this still yields a reduced-form μ_k capturing how producer prices fall with market size (selection makes the scale effect smaller when the Pareto shape parameter is large).

7.2 Intermediate-input linkages (Theorem 4)

With input–output linkages, trade policy has to distinguish between final and intermediate uses of goods to avoid “export-tax reimportation.” Theorem 4 gives first-best unilateral policies in that environment:

$$1 + s_{i,k}^{c*} = (1 + \mu_k) \frac{1 + \omega_{ii,k}^c}{1 + \bar{t}_i}, \quad (23)$$

$$1 + t_{ji,k}^{c*} = (1 + \bar{t}_i) (1 + \omega_{ji,k}^c) \frac{1 + \Lambda_{ji,k}}{1 + \omega_{ji,k}^c \Lambda_{ji,k}}, \quad (24)$$

$$1 + x_{ij,k}^{c*} = \frac{1}{1 + \omega_{ij,k}^c} \frac{1}{1 + s_{-i}^c}, \quad (25)$$

where $\Lambda_{ji,k}$ is the share of intermediate imports from j in i that are ultimately reimported by j and s_{-i}^c is a (final-good) tax shifter chosen by foreign governments. **Interpretation:** input–output linkages add a further strategic channel (reimportation) that modifies the effective ToT wedge for trade taxes, but the basic structure (ToT terms + corrective industrial term) remains.

7.3 Political economy

The paper also considers non-welfare objectives (political weights on industries). In the reading, this adds another layer of “domestic wedges” that can (i) rationalize industrial policies even without scale economies, and (ii) strengthen the case for deep agreements if political distortions interact with ToT incentives.

8 Model 5: Estimating trade elasticities and scale elasticities (Section V)

8.1 Estimating equation

Using Colombian import transactions, the paper estimates σ_k and γ_k (hence μ_k) from a demand equation that links variety-level sales, prices, and within-origin market shares. The key log relationship is (paper Eq. (15)):

$$\ln \tilde{x}_{j,kt}(\omega) = (1 - \sigma_k) \ln \tilde{p}_{j,kt}(\omega) + \left(1 - \frac{\sigma_k - 1}{\gamma_k - 1}\right) \ln \lambda_{j,kt}(\omega) + D_{kt} + \ln \phi_{j,kt}(\omega), \quad (26)$$

where D_{kt} is a product-year term common to all varieties and ϕ is a variety-level demand/quality shifter.

First-differencing gives:

$$\Delta \ln \tilde{x}_{j,kt}(\omega) = (1 - \sigma_k) \Delta \ln \tilde{p}_{j,kt}(\omega) + \left(1 - \frac{\sigma_k - 1}{\gamma_k - 1}\right) \Delta \ln \lambda_{j,kt}(\omega) + \Delta \ln \phi_{j,kt}(\omega). \quad (27)$$

8.2 Instruments and identification (core idea)

Two endogenous regressors appear: $\Delta \ln \tilde{p}$ and $\Delta \ln \lambda$. The paper uses:

- a **shift–share exchange-rate instrument** for price changes,

$$z_{j,kt}(\omega) = \sum_{m=1}^{12} s_{j,k,t-1}(\omega, m) \Delta \ln e_{j,t}(m),$$

with $s_{j,k,t-1}(\omega, m)$ the lagged monthly sales share and $e_{j,t}(m)$ the bilateral COP exchange rate (paper condition (C1)),

- plus two instruments for the within-origin share term $\Delta \ln \lambda$ (paper condition (C2)): (i) change in the number of origin- j firms selling in category k in Colombia, and (ii) change in the number of HS10 categories served by firm ω .

They use efficient GMM/2SLS with Adao–Kolesár–Morales style corrections for shift–share designs.

8.3 Table 3: estimated markups and scale elasticities (industry-level)

Table 3 reports $(\sigma_k - 1)$, the ratio $(\sigma_k - 1)/(\gamma_k - 1)$, and the implied $\mu_k = 1/(\gamma_k - 1)$. A key empirical fact is that across industries, σ_k and μ_k are **strongly negatively correlated** (the table reports $\text{cov}(\sigma_k, \mu_k) \approx -0.65$).

Reading of Table 3:

- There is substantial heterogeneity in μ_k across sectors (some sectors behave almost competitive, others have large markups/scale effects).
- The negative comovement between μ_k and σ_k is exactly the configuration that makes unilateral “markup correction” politically/economically difficult (it worsens ToT in low-trade-elasticity sectors), which foreshadows the quantitative results on deep cooperation.

TABLE 4—THE GAINS FROM NONCOOPERATIVE POLICIES AND THE CONSEQUENCES OF RETALIATION

Country	Restricted entry				Free entry			
	1st-best	2nd-best trade tax	3rd-best import tax	Post-retaliation	1st-best	2nd-best trade tax	3rd-best import tax	Post-retaliation
AUS	0.90%	0.21%	0.14%	−0.83%	2.30%	0.58%	0.35%	2.85%
AUT	1.31%	0.66%	0.45%	−1.67%	2.12%	1.11%	0.58%	−2.34%
BEL	1.31%	0.69%	0.50%	−3.07%	1.66%	0.90%	0.55%	−4.13%
BGR	2.01%	0.63%	0.52%	0.04%	5.48%	1.88%	0.83%	0.23%
BRA	1.87%	0.23%	0.17%	0.93%	4.31%	0.64%	0.36%	3.21%
CAN	1.72%	0.57%	0.45%	−1.10%	3.42%	1.19%	0.46%	0.34%
CHE	1.04%	0.36%	0.46%	−0.92%	1.28%	0.76%	0.53%	−1.26%
CHN	1.76%	0.27%	0.24%	1.43%	3.70%	0.39%	0.28%	2.90%
CYP	1.75%	0.68%	0.61%	−4.20%	5.83%	1.56%	1.43%	−10.88%
CZE	1.68%	0.88%	0.54%	−1.70%	2.86%	1.60%	0.73%	−2.13%
DEU	1.71%	0.78%	0.49%	−0.37%	2.60%	1.34%	0.65%	−0.10%
DNK	1.22%	0.38%	0.48%	−2.41%	1.94%	0.91%	0.49%	−4.78%
ESP	1.53%	0.52%	0.39%	−0.15%	2.51%	1.03%	0.49%	0.29%
EST	1.18%	0.62%	0.45%	−4.86%	2.69%	1.42%	0.56%	−7.90%
FIN	1.40%	0.54%	0.25%	−0.76%	2.00%	0.83%	0.47%	−1.08%
FRA	1.20%	0.45%	0.34%	−0.78%	2.52%	1.10%	0.50%	0.01%
GBR	1.09%	0.48%	0.42%	−0.65%	2.10%	1.05%	0.58%	0.17%
GRC	1.82%	0.59%	0.53%	−0.20%	2.75%	1.08%	0.68%	−0.04%
HRV	1.03%	0.55%	0.45%	−2.27%	1.79%	0.73%	0.52%	−2.99%
HUN	2.25%	1.06%	0.73%	−2.85%	4.08%	2.28%	0.96%	−3.33%
IDN	2.00%	0.36%	0.26%	−0.37%	4.81%	1.45%	0.50%	2.42%
IND	1.81%	0.35%	0.31%	1.60%	4.20%	1.10%	0.37%	2.82%
IRL	0.86%	0.68%	0.53%	−2.07%	1.49%	0.89%	0.40%	−3.32%
ITA	1.50%	0.46%	0.26%	0.36%	2.75%	0.95%	0.48%	0.80%
JPN	1.48%	0.32%	0.22%	0.31%	2.97%	0.75%	0.42%	1.55%
KOR	2.12%	0.65%	0.49%	0.14%	4.37%	1.62%	0.73%	1.25%
LTU	2.50%	0.93%	0.76%	−1.38%	3.54%	1.26%	0.85%	−2.47%
LUX	0.93%	0.81%	0.78%	−3.59%	1.52%	1.15%	1.01%	−4.90%
LVA	0.91%	0.54%	0.44%	−4.26%	1.57%	0.80%	0.46%	−4.85%
MEX	2.24%	0.60%	0.44%	−0.92%	4.96%	1.36%	0.76%	1.96%
MLT	1.35%	0.92%	0.82%	−3.75%	2.08%	1.39%	1.06%	−4.81%
NLD	1.35%	0.66%	0.54%	−3.44%	1.95%	0.98%	0.59%	−3.73%
NOR	1.19%	0.40%	0.27%	−0.74%	2.02%	0.81%	0.41%	−0.33%
POL	2.19%	0.76%	0.65%	−0.26%	4.03%	1.62%	0.81%	0.80%
PRT	2.04%	0.74%	0.65%	−0.36%	3.92%	1.72%	0.80%	0.73%
ROU	2.05%	0.77%	0.67%	−0.42%	3.92%	1.72%	0.98%	1.15%
RUS	2.39%	0.32%	0.27%	0.84%	5.25%	1.39%	0.37%	2.80%
SVK	2.01%	1.06%	0.79%	−2.17%	3.50%	2.09%	1.08%	−2.73%
SVN	1.42%	0.87%	0.67%	−3.09%	1.35%	1.20%	0.90%	−4.92%
SWE	1.21%	0.62%	0.45%	−0.95%	1.54%	0.77%	0.49%	−1.48%
TUR	1.45%	0.46%	0.32%	−1.52%	3.52%	1.34%	0.61%	−6.33%
TWN	2.18%	0.69%	0.56%	−0.94%	4.92%	1.85%	0.79%	0.81%
USA	1.53%	0.32%	0.27%	0.69%	3.04%	0.80%	0.30%	2.15%
Average	1.59%	0.60%	0.47%	−1.23%	3.05%	1.19%	0.63%	−1.20%

Notes: The data source is the 2014 World Input-Output Database (WIOD) (Timmer et al. 2015; WIOD 2021). The first-best policy is characterized by Theorem 1; second-best trade taxes are characterized by Theorem 2; and third-best import taxes are characterized by Theorem 3. Post-retaliation corresponds to a situation where the home country sets its first-best policies and the rest of the world retaliates.

TABLE 3—INDUSTRY-LEVEL ESTIMATION RESULTS

Sector	ISIC4 codes	Estimated parameter			Obs.	Weak Ident. Test
		$\sigma_k - 1$	$\frac{\sigma_k - 1}{\gamma_k - 1}$	μ_k		
Agriculture and Mining	100-1499	6.227 (2.345)	0.891 (0.148)	0.143 (0.059)	11,568	2.40
Food	1500-1699	2.303 (0.765)	0.905 (0.046)	0.393 (0.132)	19,615	6.27
Textiles, Leather and Footwear	1700-1999	3.359 (0.353)	0.753 (0.022)	0.224 (0.024)	125,120	66.65
Wood	2000-2099	3.896 (1.855)	0.891 (0.195)	0.229 (0.120)	5,872	1.41
Paper	2100-2299	2.646 (1.106)	0.848 (0.061)	0.320 (0.136)	37,376	3.23
Petroleum	2300-2399	0.636 (0.464)	0.776 (0.119)	1.220 (0.809)	3,973	2.83
Chemicals	2400-2499	3.966 (0.403)	0.921 (0.025)	0.232 (0.024)	133,142	38.01
Rubber and Plastic	2500-2599	5.157 (1.176)	0.721 (0.062)	0.140 (0.034)	106,398	7.16
Minerals	2600-2699	5.283 (1.667)	0.881 (0.108)	0.167 (0.056)	27,952	3.53
Basic and Fabricated Metals	2700-2899	3.004 (0.484)	0.627 (0.030)	0.209 (0.035)	153,102	20.39
Machinery	2900-3099	7.750 (1.330)	0.927 (0.072)	0.120 (0.023)	263,797	12.01
Electrical and Optical Equipment	3100-3399	1.235 (0.323)	0.682 (0.017)	0.552 (0.145)	257,775	26.27
Transport Equipment	3400-3599	2.805 (0.834)	0.363 (0.036)	0.129 (0.041)	85,920	5.50
N.E.C. and Recycling	3600-3800	6.169 (1.012)	0.938 (0.090)	0.152 (0.029)	70,264	11.57

Notes: Estimation results of equation (16). Standard errors in parentheses. The estimation is conducted with HS10 product-year fixed effects. All standard errors are simultaneously clustered by product-year and by origin-product, which is akin to the correction proposed by Aday, Kolesár, and Morales (2019). The weak identification test statistics are the F -statistics from the Kleibergen-Paap Wald test for weak identification of all instrumented variables. The test for overidentification is not reported due to the pitfalls of the standard overidentification Sargan-Hansen J -test in the multidimensional large datasets pointed by Angrist, Imbens, and Rubin (1996).

9 Quantitative results (Section VI): welfare gains and the role of retaliation/cooperation

9.1 Table 4: gains from unilateral policy and consequences of retaliation

Table 4 reports (by country) the maximum welfare gain from a unilateral deviation relative to the status quo under:

- **1st-best unilateral policy** (trade + industrial instruments, Theorem 1),
- **2nd-best trade taxes** (no industrial subsidies, Theorem 2),
- **3rd-best import tariffs** (no export subsidies and no industrial subsidies, Theorem 3),
- **post-retaliation** outcomes where trading partners respond with their own optimal trade taxes.

Results are shown for both *restricted entry* (profits) and *free entry* (scale economies).

Key takeaways take from Table 4:

- **Temptation:** the average unilateral gains are economically meaningful (especially with free entry/scale effects).
- **But retaliation is costly:** “post-retaliation” welfare changes are negative on average, consistent with trade-war losses.
- **Big countries differ:** some large countries (notably the U.S.) can remain positive even post-retaliation, which is exactly why shallow agreements are hard but crucial.

9.2 Table 5: immiserizing effects of noncoordinated industrial policies

Table 5 isolates industrial policy: it compares average welfare effects of implementing the corrective industrial subsidies unilaterally versus in a coordinated way.

Reading of Table 5:

- **Unilateral industrial policy can be welfare reducing on average** (immiserizing), especially under free entry where scale effects are larger/stronger.
- **Coordination flips the sign:** coordinated implementation delivers sizable positive gains, reinforcing the “deep cooperation” message.

TABLE 5—IMMISERIZING EFFECTS OF NONCOORDINATED INDUSTRIAL POLICIES

	Restricted entry		Free entry	
	Unilateral	Coordinated	Unilateral	Coordinated
Gains from <i>corrective</i> industrial policies	−0.32%	1.67%	−2.78%	3.42%

Notes: The data source is the 2014 World Input-Output Database (Timmer et al. 2015; WIOD 2021). The columns titled “Unilateral” report welfare gains when a country unilaterally adopts industrial subsidies that restore marginal cost pricing in the domestic economy. The columns titled “Coordinated” report welfare gains when all countries simultaneously adopt industrial subsidies that restore marginal cost pricing globally. The average gains are calculated as the simple average across all 43 countries in the WIOD sample. Country-level results are reported in online Appendix X.

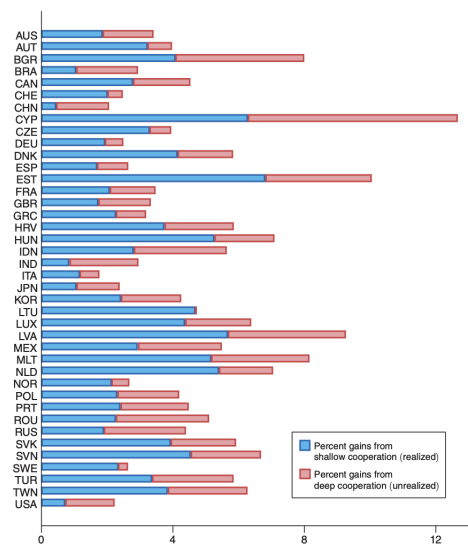


FIGURE 2. THE WELFARE GAINS FROM DEEP AND SHALLOW COOPERATION

Notes: The data source is the 2014 World Input-Output Database (Timmer et al. 2015; WIOD 2021). Welfare effects are computed under restricted entry. The gains from deep cooperation correspond to welfare gains when moving from the status quo to the globally efficient equilibrium in which all countries coordinate their corrective Pigouvian subsidies. The gains from shallow cooperation correspond to the avoided welfare losses when moving from the status quo to a noncooperative equilibrium where all countries adopt Nash trade taxes and subsidies.

9.3 Figure 2: realized (shallow) vs unrealized (deep) gains from cooperation

Figure 2 decomposes the gains from cooperation into:

- **shallow cooperation:** avoided welfare losses from a trade war (tariffs/export subsidies),
- **deep cooperation:** additional gains from coordinating industrial policy (correcting markups/scale distortions).

9.4 Figure 3: why countries might support deep cooperation

Figure 3 plots prospective gains from deep cooperation (x-axis) against maximal unilateral gains (y-axis). If a point is below the 45-degree line, deep cooperation dominates the best unilateral deviation.

Takeaway from Figures 2–3:

- Many countries have more to gain from deep cooperation than from any unilateral deviation, so there is a potential “coalition for deep agreements” even when unilateral ToT incentives exist.

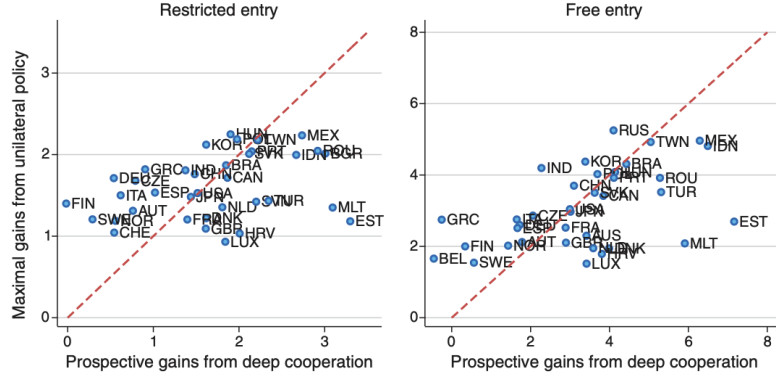


FIGURE 3. DEEP COOPERATION VERSUS UNILATERAL POLICY INTERVENTIONS

Notes: The data source is the 2014 World Input-Output Database (Timmer et al. 2015; WIOD 2021). The gains from first-best noncooperative policy are the gains when each country implements the policy characterized by Theorem 1 and the rest of the world is passive. The gains from global cooperation correspond to a scenario where all countries forgo trade taxation and apply industrial subsidies that restore marginal cost pricing.

- The negative empirical $\text{cov}(\mu_k, \sigma_k)$ makes deep cooperation particularly valuable: the ToT cost of unilateral industrial policy is too high in exactly the sectors where correction is most needed.

10 Short conclusion

The paper’s punchline is that heterogeneity in profits/markups and scale economies fundamentally changes trade-policy design: efficient policy is mainly *industrial* (correct domestic wedges), while unilateral incentives push governments toward ToT manipulation and away from corrective subsidies. Empirically, because industries with high scale/markup distortions tend to have low trade elasticities, unilateral industrial policies can be immiserizing and are hard to sustain. This creates a strong rationale for **deep** international agreements that coordinate industrial policy in addition to regulating trade taxes.

11 Conclusion

11.1 Summary

The paper studies optimal trade and industrial policy in quantitative trade models where industries differ in *profit-generating markups* (restricted entry) and/or *scale economies* (free entry). These features generate an additional, **within-country intersectoral misallocation wedge** that is conceptually distinct from the classic **terms-of-trade (ToT) wedge** emphasized in traditional trade-policy theory.

The main conclusions are:

1. **Trade restrictions are an ineffective second-best instrument for domestic misallocation.** Even when designed optimally, trade taxes cannot replicate more than a fraction of the welfare gains attainable under the first-best policy mix that uses *Pigouvian industrial subsidies* to correct the domestic markup/scale distortion.
2. **Unilateral corrective industrial policies can be self-defeating (immiserizing).** Correcting the domestic distortion raises output in high-returns-to-scale / high-markup industries,

but this expansion can worsen the terms of trade. In the empirically relevant configuration, the ToT deterioration is large enough that unilateral “correction” can reduce real income.

3. **Deep international agreements (industrial-policy coordination) unlock large welfare gains.** If countries can *coordinate* corrective industrial subsidies internationally (reciprocal implementation), the ToT externality is neutralized, and the domestic misallocation correction becomes strongly welfare-improving.
4. **Shallow cooperation (tariff discipline) is not enough when ToT concerns block industrial correction.** Shallow agreements can prevent costly trade wars, but they do not, by themselves, ensure that countries implement the industrial policies needed to eliminate sectoral misallocation. Hence a “deep” agreement can deliver additional (unrealized) gains.

11.2 Quantitative punchlines

Using micro trade data to estimate industry-level elasticities and a multi-country input–output quantitative model (WIOD), the paper reports several salient quantitative findings:

- **Second-best trade taxes replicate only about one-third of the first-best gains.** Under free entry, the *first-best noncooperative* policy (trade taxes + corrective subsidies) increases welfare by about $\approx 3.05\%$ on average, while *second-best trade taxes* (without industrial subsidies) deliver only $\approx 1.19\%$. Import tariffs alone are roughly half as effective as trade taxes that also allow export subsidies.
- **Retaliation overturns unilateral trade-policy gains.** When trading partners reciprocally adopt their own optimal trade taxes, the *net* effect becomes a welfare *loss* of about $\approx 1.20\%$ on average (free entry case), consistent with large trade-war losses.
- **Unilateral industrial correction is immiserizing; coordinated correction is strongly positive.** A unilateral move to “restore marginal cost pricing” through corrective industrial subsidies reduces real income by about $\approx 0.32\%$ (restricted entry) and $\approx 2.78\%$ (free entry) for the average country, whereas *coordinated* implementation raises welfare by about $\approx 1.67\%$ (restricted entry) and $\approx 3.42\%$ (free entry).
- **Deep cooperation delivers additional gains beyond shallow cooperation.** Avoided welfare losses from a global move to noncooperative trade taxes (“shallow cooperation” gains) are large on average (about $\approx 3.2\%$ under restricted entry), while coordinated industrial-policy implementation adds further gains (about $\approx 1.6\%$ under restricted entry, larger under free entry).

11.3 Economic intuition (why these results arise)

The mechanism is a clean second-best logic with two wedges:

1. **A domestic (within-country) wedge:** markups/scale economies imply that private marginal costs do not equal social marginal costs within an industry. Denote the key reduced-form distortion parameter by μ_k . The globally efficient remedy is a **Pigouvian output subsidy** that restores marginal-cost pricing:

$$1 + s_{i,k}^* = 1 + \mu_k, \quad t_{ji,k}^* = x_{ij,k}^* = 0.$$

Thus, in the cooperative benchmark, industrial policy (not trade policy) is the natural instrument.

2. **A terms-of-trade wedge:** unilateral governments care about how policy affects export prices and import prices. Corrective subsidies expand output and exports in targeted sectors, which can *worsen* the terms of trade.
3. **Why trade taxes perform poorly as substitutes:** trade taxes mainly operate by distorting the *across-country* allocation (imports vs. domestic), not by directly repairing the *within-country* intersectoral wedge. When governments try to use tariffs/export subsidies to mimic industrial correction, the ToT motive pushes the policy in the opposite direction, creating a fundamental “ToT vs. allocative efficiency” conflict.
4. **Why unilateral industrial policy can be immiserizing:** the paper estimates that industries with larger μ_k tend to have *lower* trade elasticities σ_k , i.e. $\text{cov}(\sigma_k, \mu_k) < 0$. Intuitively, the sectors that “need” correction most are also the sectors in which expanding exports has the largest ToT cost, so unilateral correction can reduce welfare even though it fixes a domestic wedge.
5. **Why deep agreements matter:** if countries coordinate industrial subsidies reciprocally, the ToT externality is internalized. Then the Pigouvian logic dominates, and correcting domestic misallocation becomes strongly welfare-improving.

11.4 Takeaway

With heterogeneous markups/scale effects across industries, misallocation is primarily a domestic wedge best corrected by industrial subsidies, but unilateral governments underprovide (or reverse) such correction due to terms-of-trade losses, so trade taxes are weak substitutes and deep international coordination of industrial policy is required to realize large welfare gains.